

MPSS.HI.4.0 LTE Higher Order RxD

Application Note

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Qualcomm Technologies, Inc.
5775 Morehouse Drive
San Diego, CA 92121
U.S.A.

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linyz4@motorola.com

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1 Introduction

This document introduces higher order receive diversity (HORxD). It shows how to control primary and secondary Rx chains on LTE on the MPSS.HI.4.0 modem.

1.1 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `cp armcc armcpp`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, `copy a:*. * b:.`

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

1.2 Technical assistance

For assistance or clarification on information in this document, submit a case to Qualcomm Technologies, Inc. (QTI) at <https://createpoint.qti.qualcomm.com/>.

If you do not have access to the CDMATech Support website, register for access or send email to support.cdmatech@qti.qualcomm.com.

It is important to correctly file RF cases with the proper problem area code. This helps to ensure that the case assignment is sent to the correct technology support team for prompt response.

Log cases with the following codes.

Initial problem type	Software
Problem area 1	RF software
Problem area 2	Antenna software
Problem area 3	Single chain antenna test

2 HORxD overview

2.1 HORxD introduction

In wireless communication, diversity gain can be achieved by time diversity, frequency diversity, spatial diversity, and so on. Higher order receive diversity (HORxD) is a type of spatial diversity that uses additional antennas on the user equipment (UE) for Rx.

MPSS.HI.4.0 supports up to four Rx antennas for LTE.

Trade-offs include:

- Pros – Improves performance, user experience, and throughput
- Cons – Increases antenna size/number, component count, and power required

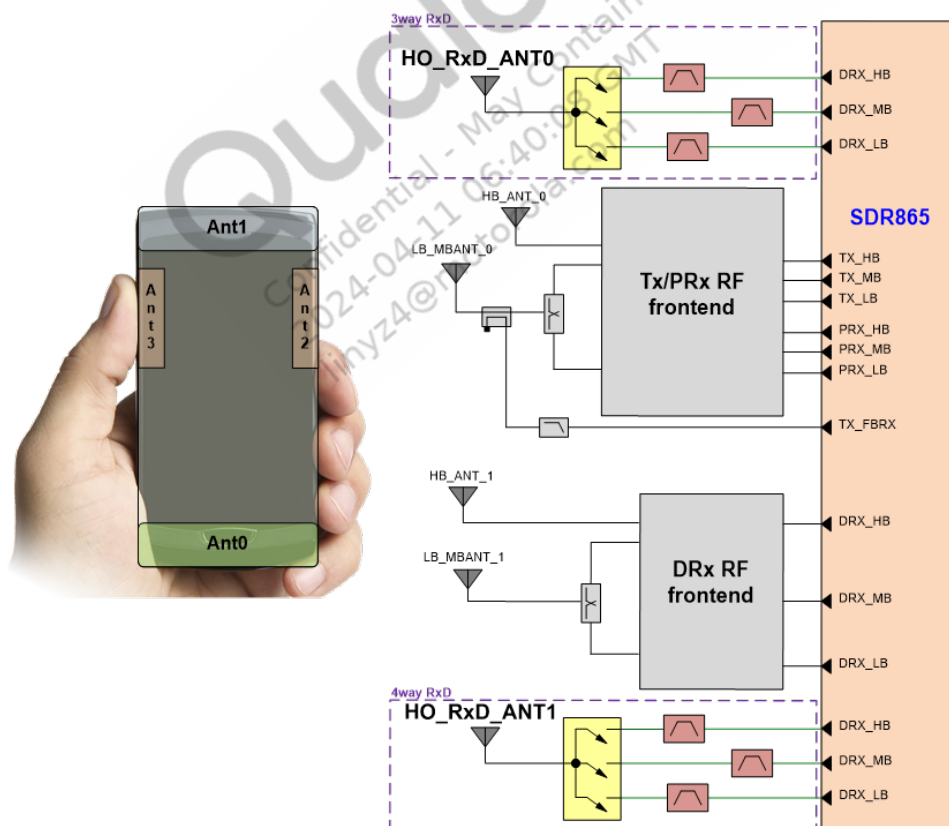


Figure 2-1 HORxD hardware design diagram

HORxD has the following benefits:

- This UE feature does not require any network upgrade or intervention.
- This feature boosts performance compared to legacy 2Rx mode:
 - Throughput gains
 - DL network capacity
 - Improves spectrum efficiency and cell edge performance
 - Improves Rx sensitivity
- HORxD + 2DLCA/3DLCA is supported to provide performance improvement by combining the benefits of HORxD and 2/3DLCA.
- Even though HORxD increases by about 20 mA, the ARD algorithm reduces this consumption as much as possible in various network scenarios.

2.2 Adaptive Rx diversity (ARD) overview

ARD refers to the management of the number of receive chains. The goal is to optimize the power-performance tradeoff. There are many state transitions that map different chain settings to adapt all kinds of NW scenarios.

ARD introduces the following new features in the MPSS.HI.4.0 modem:

- Reduce CRC failures in 1Rx.
 - Add force R1 macro state and report rank 1. 1Rx is enough to decode small 1-layer grants.
 - UE goes back to steady state upon receiving any DL grant.
- Produce consistent cell status feedback (CSF) reports.
 - The CSF report is consistent within the same macro state to avoid reported rank fluctuation due to Rx mode change.
 - The RI/CQI report is not affected by a temporary fallback to 1Rx.

2.3 4X4 MIMO/HORxD/ARD differences

Usually, 4 standalone Rx path RF hardware designs are used for the 4X4 MIMO feature, and 4X4 MIMO is activated by NW.

In the MIMO inactive scenario, the UE will enter HORxD mode, which is defined by QTI and run by the UE's algorithm. The HORxD capability is neither reported to NW nor controlled by NW.

When HORxD runs, it introduces more power consumption. To reduce the power impact, ARD is introduced, which dynamically adjusts the chain setting based on different radio link conditions.

2.4 EFS configuration for 4X4MIMO/HORxD/ARD

By default, 4X4 MIMO (if RF already supports), HORxD, and ARD are enabled on the LTE protocol side. The following embedded file system (EFS) configurations disable these features.

Disable ARD

When ARD is disabled, all bands are forced to work in normal 2Rx (Rx0+Rx1).

EFS file – *ard_dbg*

Location – */nv/item_files/modem/lte/ML1/*

- Set the first 10 bytes to 00 00 00 00 00 00 00 04 00 FF FF FF FF 00 02 00
- Set the remaining bytes to 00

Disable 4X4 MIMO/HORxD

EFS file – *disable_cap_ies*

Location – */nv/item_files/modem/lte/rrc/cap/sable*

The following settings assume that the starting bit is bit 0:

- 4X4 MIMO – Set bits 16, 17, 21, and 23 to 1
- HORxD (single carrier) – Set bit 14 to 1
- HORxD + CA – Set bits 15 and 24 to 1

See *Enabling or Disabling LTE Features* (80-NV396-46) for details.

3 Configure LTE Rx chains

When LTE calls Rx tune, by default, it requests the Rx driver to turn on both Rx0 and Rx1. Under certain specific test setups, it may be appropriate that one of the two chains (or four chains in the ARD case) is active and the other is inactive. There are multiple ways to control the primary and diversity path of the UE. The following methods are used to select a single chain or all Rx chains to perform sensitivity measurements. To verify the command functionality, it is recommended to perform power sweep and a sensitivity test.

3.1 Use EFS file

To control a specific Rx chain, place an EFS file with filename rx_select into the UE EFS folders under the following directory:

`/nv/item_files/modem/lte/ML1/`

The file contains 1 byte. The following table lists the structure.

Table 3-1 EFS file configuration values

Bit 7 – 4 (CC info)	Rx3 (bit 3)	Rx2 (bit 2)	Rx1 (bit 1)	Rx0 (bit 0)
0000 for PCC 0001 for SCC1 0010 for SCC2 0011 for SCC3 0100 for SCC4 0101 for SCC5 0110 for SCC6	0 – Disable Rx3 1 – Enable Rx3	0 – Disable Rx2 1 – Enable Rx2	0 – Disable Rx1 1 – Enable Rx1	0 – Disable Rx0 1 – Enable Rx0

If the EFS file exists, the UE reads it at power-up.

The software can enable one Rx chain, both Rx0/Rx1 (2Rx), or all Rx0/Rx1/Rx2/Rx3 (4Rx). The valid values for bit <3:0> are 1, 2, 3, 4, 8, 15. If other values are configured, there is no action. The following lists test steps using an EFS file.

The following steps show how to test with the rx_select EFS file.

1. Load EFS into devices and reboot.
2. Start online call.
3. Single chain is triggered automatically based on the EFS file configuration.
4. Run single chain RF test.
5. Hand over to another channel/band.
6. Single chain state remains the same.
7. Run single chain RF test

8. End call.

The following tables summarize config_number for different scenarios.

Table 3-2 config_number configuration – PCC/Standalone

config_number value	Description	CC ID Bit 7:4	Rx3 Bit 3	Rx2 Bit 2	Rx1 Bit 1	Rx0 Bit 0
1	Enables Rx0 only	0	0	0	0	1
2	Enables Rx1 only	0	0	0	1	0
3	Enables Rx0 and Rx1	0	0	0	1	1
4	Enables Rx2 only	0	0	1	0	0
8	Enables Rx3 only	0	1	0	0	0
15	Enables all chains	0	1	1	1	1

Table 3-3 config_number configuration – SCC1

config_number value	Description	CC ID Bit 7:4	Rx3 Bit 3	Rx2 Bit 2	Rx1 Bit 1	Rx0 Bit 0
17	Enables Rx0 only	1	0	0	0	1
18	Enables Rx1 only	1	0	0	1	0
19	Enables Rx0 and Rx1	1	0	0	1	1
20	Enables Rx2 only	1	0	1	0	0
24	Enables Rx3 only	1	1	0	0	0
31	Enables all chains	1	1	1	1	1

Table 3-4 config_number configuration – SCC2

config_number value	Description	CC ID Bit 7:4	Rx3 Bit 3	Rx2 Bit 2	Rx1 Bit 1	Rx0 Bit 0
33	Enables Rx0 only	2	0	0	0	1
34	Enables Rx1 only	2	0	0	1	0
35	Enables Rx0 and Rx1	2	0	0	1	1
36	Enables Rx2 only	2	0	1	0	0
40	Enables Rx3 only	2	1	0	0	0
47	Enables all chains	2	1	1	1	1

Table 3-5 config_number configuration – SCC3

config_number value	Description	CC ID Bit 7:4	Rx3 Bit 3	Rx2 Bit 2	Rx1 Bit 1	Rx0 Bit 0
49	Enables Rx0 only	3	0	0	0	1
50	Enables Rx1 only	3	0	0	1	0
51	Enables Rx0 and Rx1	3	0	0	1	1
52	Enables Rx2 only	3	0	1	0	0
56	Enables Rx3 only	3	1	0	0	0
63	Enables all chains	3	1	1	1	1

Table 3-6 config_number configuration – SCC4

config_number value	Description	CC ID Bit 7:4	Rx3 Bit 3	Rx2 Bit 2	Rx1 Bit 1	Rx0 Bit 0
65	Enables Rx0 only	4	0	0	0	1
66	Enables Rx1 only	4	0	0	1	0
67	Enables Rx0 and Rx1	4	0	0	1	1
68	Enables Rx2 only	4	0	1	0	0
72	Enables Rx3 only	4	1	0	0	0
79	Enables all chains	4	1	1	1	1

Table 3-7 config_number configuration – SCC5

config_number value	Description	CC ID Bit 7:4	Rx3 Bit 3	Rx2 Bit 2	Rx1 Bit 1	Rx0 Bit 0
81	Enables Rx0 only	5	0	0	0	1
82	Enables Rx1 only	5	0	0	1	0
83	Enables Rx0 and Rx1	5	0	0	1	1
84	Enables Rx2 only	5	0	1	0	0
88	Enables Rx3 only	5	1	0	0	0
95	Enables all chains	5	1	1	1	1

Table 3-8 config_number configuration– SCC6

config_number value	Description	CC ID Bit 7:4	Rx3 Bit 3	Rx2 Bit 2	Rx1 Bit 1	Rx0 Bit 0
97	Enables Rx0 only	6	0	0	0	1
98	Enables Rx1 only	6	0	0	1	0
99	Enables Rx0 and Rx1	6	0	0	1	1
100	Enables Rx2 only	6	0	1	0	0
104	Enables Rx3 only	6	1	0	0	0
111	Enables all chains	6	1	1	1	1

3.2 Use the DIAG command

When LTE sends an Rx tuning request to the RF driver, the DIAG command can be sent from the QXDM Professional™ Tool (QXDM) to the protocol stack to enable or disable a specific chain. These commands are used to select a single chain, or all four chains in the ARD case, to perform sensitivity measurements.

NOTE: The DIAG command can only be sent when the RF driver Rx is tuned.

Use the following DIAG command format in the QXDM DIAG command box:

```
send_data 75 68 9 64 <8-bit config_number>
```

This command (Command ID: 0x4000) is routed to the subsystem DIAG_SUB_SYS_LTE (68). The tool_chain_code is a 1-byte value that includes the following:

- Command description

This DIAG command enables or disables a specific Rx chain while the UE is in LTE mode and camped on a specific system.

- Command payload

The payload to this DIAG command is 1 byte. The valid value is the same as the EFS file, which is listed in [Table 3-3](#) to [Table 3-8](#).

The following steps show the test with the rx_select QXDM command.

1. Issue a new initialization command during idle state:

```
send_data 75 68 9 64 128
```

2. Start online call.

3. Issue Rx_sel command after 50 ms:

```
send_data 75 68 9 64 x
```

4. Run single chain RF test.

5. Hand over to another channel/band.

6. Issue Rx_sel command again 50 ms later after reconnecting:

```
send_data 75 68 9 64 x
```

7. Run single chain RF test.
8. End call.

3.3 Set up the environment

The device must be well calibrated before test. It's suggested to use a power splitter to perform a single chain test. The following figure shows the connection topology.

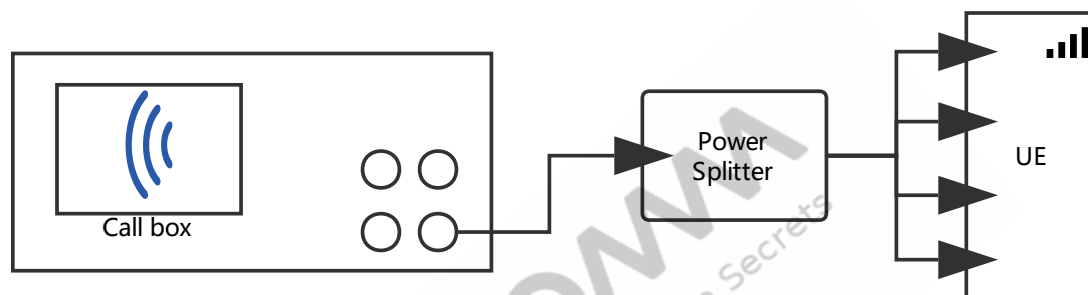


Figure 3-1 Reference connection for Rx chain test

3.4 Confirm Rx chain from log

The 0xB193 log packet can be used to check whether the QXDM command is working on software. The RSSI, RSRP, RSRQ, and FTL SNR for disabled chains show the floored values.

3.4.1 Use EFS file

The rx_sel EFS file is loaded to the phone during power-up. The corresponding configuration always takes effect in connected mode.

```
//configuration from efs take effect
[0009/01/1] QTRACE 00:05:47.515
9/1/Medium/LML1 [lte_mll_rfmgr_rx_div.c 713] RX SEL: try to
apply target selection 0x1 on cc 0, cur 0xF pending flag 0
[0009/01/1] QTRACE 00:05:47.515
9/1/Medium/LML1 [lte_mll_rfmgr_rx_div.c 661] RX SEL: fw_sleep 0
rx_mask 3 selection 0x1 on cc 0 4Rx capable 0
[0009/01/2] QTRACE 00:05:47.515
9/1/High/LML1 [lte_mll_rfmgr_rx_div.c 922] RX SEL->ARD:
LTE_ML1_DLM_ARD_RX_SEL_IND cc_id 0, num_rx 1, rx_map [0 0 0 0]
//Rx0 enabled from log packet 0xB193
1980 Jan 6 00:05:47.593 [DB] 0xB193 LTE ML1 Serving Cell Meas Response
Subscription ID = 1
Version = 1
Number of SubPackets = 1
SubPacket ID = 25
Serving Cell Measurement Result
```

```

Version = 40
SubPacket Size = 540 bytes
E-ARFCN = 300
Num of Cell = 1
Valid Rx = RX0
Cells[0]
    Physical Cell ID = 0
    Serving Cell Index = PCell
    Is Serving Cell = 1
    Current SFN = 177
    Current Subframe Number = 3
    Is Restricted = false
    Cell Timing[0] = 36994
    Cell Timing[1] = 36994
    Cell Timing SFN[0] = 177
    Cell Timing SFN[1] = 177
    Inst RSRP Rx[0] = -82.56 dBm
    Pathloss RSRP Rx[0] = -82.56 dBm
    Inst RSRP Rx[1] = NA
    Pathloss RSRP Rx[1] = NA
    Inst RSRP Rx[2] = NA
    Pathloss RSRP Rx[2] = NA
    Pathloss RSRP Rx[3] = NA
    Inst RSRP Rx[3] = NA
    Inst Measured RSRP = -81.56 dBm
    True Inst Measured RSRP = -81.56 dBm
    Filtered RSRP = -81.56 dBm
    Inst RSRQ Rx[0] = -6.50 dB
    True Inst RSRQ Rx[0] = -6.50 dB
    Inst RSRQ Rx[1] = NA
    True Inst RSRQ Rx[1] = NA
    Inst RSRQ Rx[2] = NA
    Inst RSRQ Rx[3] = NA
    Inst RSRQ = -6.50 dB
    True Inst RSRQ = -6.50 dB
    Filtered RSRQ = -6.50 dB
    Inst RSSI Rx[0] = -59.06 dBm
    Inst RSSI Rx[1] = NA
    Inst RSSI Rx[2] = NA
    Inst RSSI Rx[3] = NA
    Inst RSSI = -57.00 dBm
    DVGA Gain Log[0] = -18
    DVGA Gain Log[1] = -19
    LNA Gain Log[0] = 35
    LNA Gain Log[1] = 33
    CQI Web = 0
    Residual Frequency Error = 30440
    FTL SNR Linear Rx[0] = 110061830

```

```

FTL SNR Linear Rx[1]      = 0
FTL SNR Rx[0]             = 30.00 dB
FTL SNR Rx[1]             = NA
FTL SNR Rx[2]             = NA
FTL SNR Rx[3]             = NA
RS SNR Linear Rx[0]       = 393896
RS SNR Linear Rx[1]       = 0
RS SNR Rx[0]              = 30.00 dB
RS SNR Rx[1]              = NA
RS SNR Rx[2]              = NA
RS SNR Rx[3]              = NA
Projected Sir              = 0 dB
Post Ic Rsrq              = 4294967290 dB
CINR RX 0                  = 8151
CINR RX 1                  = NA
CINR RX 2                  = NA
CINR RX 3                  = NA

```

3.4.2 Use the DIAG command

On MPSS.HI.4.0, issue the init command in idle state. Other commands do not take effect if the init command is not sent before connecting.

```

//issue init command in idle state
[0009/01/1] QTRACE 00:27:22.711
9/1/Medium/LML1 [lte_ml1_rfmgr_rx_div.c 1082] RX SEL <- DIAG:
raw action_code=0x80, cc_id=0, selection=0
[0009/01/1] QTRACE 00:27:22.711
9/1/Medium/LML1 [lte_ml1_rfmgr_rx_div.c 1086] RX SEL: Init cmd
is received
[0009/01/2] QTRACE 00:27:22.711
9/1/High/LML1 [lte_ml1_rfmgr_rx_div.c 952] RX SEL->ARD:
LTE_ML1_DLM_ARD_MISSION_MODE_IND
//default Rx chain enabled after init command
1980 Jan 6 00:27:43.413 [6E] 0xB193 LTE ML1 Serving Cell Meas Response
Subscription ID = 1
Version = 1
Number of SubPackets = 1
SubPacket ID = 25
Serving Cell Measurement Result
  Version = 40
  SubPacket Size = 540 bytes
  E-ARFCN = 300
  Num of Cell = 1
  Valid Rx = RX0_RX1
  Cells[0]
    Physical Cell ID = 0
    Serving Cell Index = PCell
    Is Serving Cell = 1

```

Current SFN	= 391
Current Subframe Number	= 6
Is Restricted	= false
Cell Timing[0]	= 103370
Cell Timing[1]	= 103370
Cell Timing SFN[0]	= 391
Cell Timing SFN[1]	= 391
Inst RSRP Rx[0]	= -82.88 dBm
Pathloss RSRP Rx[0]	= -82.63 dBm
Inst RSRP Rx[1]	= -92.50 dBm
Pathloss RSRP Rx[1]	= -92.56 dBm
Inst RSRP Rx[2]	= NA
Pathloss RSRP Rx[2]	= NA
Pathloss RSRP Rx[3]	= NA
Inst RSRP Rx[3]	= NA
Inst Measured RSRP	= -82.88 dBm
True Inst Measured RSRP	= -82.88 dBm
Filtered RSRP	= -82.63 dBm
Inst RSRQ Rx[0]	= -7.63 dB
True Inst RSRQ Rx[0]	= -7.63 dB
Inst RSRQ Rx[1]	= -10.56 dB
True Inst RSRQ Rx[1]	= -10.56 dB
Inst RSRQ Rx[2]	= NA
Inst RSRQ Rx[3]	= NA
Inst RSRQ	= -7.63 dB
True Inst RSRQ	= -7.63 dB
Filtered RSRQ	= -7.44 dB
Inst RSSI Rx[0]	= -58.25 dBm
Inst RSSI Rx[1]	= -64.94 dBm
Inst RSSI Rx[2]	= NA
Inst RSSI Rx[3]	= NA
Inst RSSI	= -58.25 dBm
DVGA Gain Log[0]	= -17
DVGA Gain Log[1]	= -8
LNA Gain Log[0]	= 35
LNA Gain Log[1]	= 53
CQI Web	= 0
Residual Frequency Error	= 32767
FTL SNR Linear Rx[0]	= 148036784
FTL SNR Linear Rx[1]	= 989220
FTL SNR Rx[0]	= 30.00 dB
FTL SNR Rx[1]	= 11.80 dB
FTL SNR Rx[2]	= NA
FTL SNR Rx[3]	= NA
RS SNR Linear Rx[0]	= 593294
RS SNR Linear Rx[1]	= 5396
RS SNR Rx[0]	= 30.00 dB
RS SNR Rx[1]	= 13.30 dB

```

RS SNR Rx[2]           = NA
RS SNR Rx[3]           = NA
Projected Sir           = 0 dB
Post Ic Rsrq           = 4294967289 dB
CINR RX 0               = 8614
CINR RX 1               = 3426
CINR RX 2               = NA
CINR RX 3               = NA

//send command send_data 75 68 9 64 15 in connected mode
[0009/01/1]             QTRACE                      00:27:54.080
9/1/Medium/LML1         [lte_ml1_rfmgr_rx_div.c     1082] RX SEL <- DIAG:
raw action_code=0xF, cc_id=0, selection=15
[0009/01/1]             QTRACE                      00:27:54.080
9/1/Medium/LML1         [lte_ml1_rfmgr_rx_div.c     661] RX SEL: fw_sleep 0
rx_mask 15 selection 0xF on cc 0 4Rx capable 1
[0009/01/2]             QTRACE                      00:27:54.080
9/1/High/LML1           [lte_ml1_rfmgr_rx_div.c     922] RX SEL->ARD:
LTE_ML1_DLM_ARD_RX_SEL_IND cc_id 0, num_rx 4, rx_map [0 1 2 3]
//4Rx enabled from log packet 0xB193
1980 Jan  6  00:27:54.093 [7A]  0xB193  LTE ML1 Serving Cell Meas Response
Subscription ID = 1
Version                = 1
Number of SubPackets = 1
SubPacket ID           = 25
Serving Cell Measurement Result
  Version = 40
  SubPacket Size = 540 bytes
  E-ARFCN           = 300
  Num of Cell        = 1
  Valid Rx           = RX0_RX1_RX2_RX3
  Cells[0]
    Physical Cell ID      = 0
    Serving Cell Index    = PCell
    Is Serving Cell       = 1
    Current SFN           = 435
    Current Subframe Number = 6
    Is Restricted         = false
    Cell Timing[0]        = 103370
    Cell Timing[1]        = 103370
    Cell Timing SFN[0]     = 435
    Cell Timing SFN[1]     = 435
    Inst RSRP Rx[0]       = -82.75 dBm
    Pathloss RSRP Rx[0]   = -82.81 dBm
    Inst RSRP Rx[1]       = -92.50 dBm
    Pathloss RSRP Rx[1]   = -92.50 dBm
    Inst RSRP Rx[2]       = -77.94 dBm
    Pathloss RSRP Rx[2]   = -77.94 dBm
    Pathloss RSRP Rx[3]   = -80.00 dBm

```

Inst RSRP Rx[3]	= -80.00 dBm
Inst Measured RSRP	= -82.75 dBm
True Inst Measured RSRP	= -82.75 dBm
Filtered RSRP	= -82.81 dBm
Inst RSRQ Rx[0]	= -10.31 dB
True Inst RSRQ Rx[0]	= -10.31 dB
Inst RSRQ Rx[1]	= -10.63 dB
True Inst RSRQ Rx[1]	= -10.63 dB
Inst RSRQ Rx[2]	= -8.25 dB
Inst RSRQ Rx[3]	= -9.25 dB
Inst RSRQ	= -10.31 dB
True Inst RSRQ	= -10.31 dB
Filtered RSRQ	= -10.31 dB
Inst RSSI Rx[0]	= -55.50 dBm
Inst RSSI Rx[1]	= -64.88 dBm
Inst RSSI Rx[2]	= -52.69 dBm
Inst RSSI Rx[3]	= -53.81 dBm
Inst RSSI	= -55.50 dBm
DVGA Gain Log[0]	= -17
DVGA Gain Log[1]	= -8
LNA Gain Log[0]	= 35
LNA Gain Log[1]	= 53
CQI Web	= 0
Residual Frequency Error	= 32767
FTL SNR Linear Rx[0]	= 122647815
FTL SNR Linear Rx[1]	= 1642883
FTL SNR Rx[0]	= 30.00 dB
FTL SNR Rx[1]	= 14.00 dB
FTL SNR Rx[2]	= 30.00 dB
FTL SNR Rx[3]	= 30.00 dB
RS SNR Linear Rx[0]	= 582928
RS SNR Linear Rx[1]	= 4822
RS SNR Rx[0]	= 30.00 dB
RS SNR Rx[1]	= 12.80 dB
RS SNR Rx[2]	= 29.50 dB
RS SNR Rx[3]	= 28.70 dB
Projected Sir	= 0 dB
Post Ic Rsrq	= 4294967286 dB
CINR RX 0	= 8606
CINR RX 1	= 3262
CINR RX 2	= 7535
CINR RX 3	= 7306

3.5 Limitations

The provided methods can be used for an RF/antenna single chain test, but they are only for test. Ensure that all the chains are enabled on commercial devices.

There are some limitations on controlling Rx chains:

- Use a call box with a single SIM to perform the test.
- No measurement is configured.
- No CDRX is configured.
- No intra noncontinuous 4+2 case.
- PCell with 4Rx while SCell has 2Rx for intra noncontinuous CA.
- SCells should always be activated during the test if CA is configured.
- There is no support for an unlicensed band.

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linyz4@motorola.com

A References

A.1 Related documents

Title	Number
Qualcomm Technologies, Inc.	
Enabling or Disabling LTE Features	80-NV396-46
MPSS.HI.1.0 RF Software Overview	80-PM475-1

A.2 Acronyms and terms

Acronym or term	Definition
ARD	Adaptive Rx diversity
CSF	Cell status feedback
EFS	Embedded file system
HORxD	Higher order receive diversity
UE	User equipment